



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ANISH NAIR

PRN: 25070122037

BATCH: A2

ASSIGNMENT No :2

TITLE: Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Assignment 2: Methods and Static Members

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Outcome

VS code Folder stuff > J as2student.java

```
1 public class as2student{
2     String name;
3     int rollNo;
4
5     // Default constructor
6     as2student() {
7         name = "Unknown";
8         rollNo = 0;
9     }
10
11    // Parameterized constructor
12    as2student(String name, int rollNo) {
13        this.name = name;
14        this.rollNo = rollNo;
15    }
16
17    void display() {
18        System.out.println("Name: " + name + ", Roll No: " + rollNo);
19    }
20
21    public static void main(String[] args) {
22        as2student s1 = new as2student(); // uses default constructor
23        as2student s2 = new as2student("Anish", 21); // uses parameterized constructor
24
25        s1.display();
26        s2.display();
27    }
28 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Work stuff\VS code stuff\VS code Folder stuff> cd "c:\Work stuff\VS code stuff\VS c
Folder stuff\" ; if ($?) { javac as2student.java } ; if ($?) { java as2student }
Name: Unknown, Roll No: 0
Name: Anish, Roll No: 21
PS C:\Work stuff\VS code stuff\VS code Folder stuff> □
```

```

VS code Folder stuff > J hw.java
1  public class hw {
2      String brand;
3      String model;
4      double price;
5
6      // Default constructor
7      hw() {
8          brand = "N/A";
9          model = "N/A";
10         price = 0.0;
11     }
12
13     // Parameterized constructor
14     hw(String brand, String model, double price) {
15         this.brand = brand;
16         this.model = model;
17         this.price = price;
18     }
19
20     // Copy constructor
21     hw(hw m) {
22         this.brand = m.brand;
23         this.model = m.model;
24         this.price = m.price;
25     }
26
27     void display() {
28         System.out.println("Brand: " + brand + ", Model: " + model + ", Price: " + price);
29     }
30
31     public static void main(String[] args) {
32         hw m1 = new hw(); // default
33         hw m2 = new hw("Samsung", "Galaxy S23", 69999.0); // parameterized
34         hw m3 = new hw(m2); // copy of m2
35
36         m1.display();
37         m2.display();
38         m3.display();
39     }
40 }

```

```

Brand: N/A, Model: N/A, Price: 0.0
Brand: Samsung, Model: Galaxy S23, Price: 69999.0
Brand: Samsung, Model: Galaxy S23, Price: 69999.0
PS C:\Work stuff\VS code stuff\VS code Folder stuff> 

```

Conclusion: Constructors (default, parameterized, and copy) were successfully implemented to initialize and manage objects for the Student and Mobile Phone Inventory classes.

This demonstrated how different constructor types simplify object creation and enable efficient data duplication in Java.

Exercise

1. Create a **Student class** using default and parameterized constructors to initialize student name and roll number.

2. Develop a **Mobile Phone Inventory System** using different constructors to initialize mobile details and create duplicate object records.